

INDUSTRIAL AC INDUCTION MOTOR — MAINTENANCE & TROUBLESHOOTING MANUAL

Equipment: Three-Phase Squirrel-Cage Induction Motor

Model (simulated): SIM-MTR-01 | Power: 15 kW | Voltage: 415V, 3-phase | Speed: 1450 RPM

Document for academic/demo use — simulated non-critical equipment

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SECTION 1: EQUIPMENT OVERVIEW

The SIM-MTR-01 is a standard TEFC (Totally Enclosed Fan-Cooled) three-phase induction motor used to drive pumps, fans, conveyors, and compressors in general industrial settings. It consists of a stator winding, a squirrel-cage rotor, two end-shield bearings, and an external cooling fan.

Key components:

- Stator windings (insulated copper coils)
- Rotor and shaft assembly
- Drive-end (DE) and non-drive-end (NDE) bearings
- Terminal box with connection studs
- Cooling fan and fan cover
- Nameplate (rating, frame size, insulation class)

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SECTION 2: SAFETY PRECAUTIONS

WARNING: All maintenance must be performed by trained personnel only.

This is decision-support documentation, not a substitute for site-specific lockout/tagout (LOTO) procedures or manufacturer instructions.

Before ANY inspection or maintenance:

1. Stop the motor and isolate the electrical supply at the breaker/isolator.
2. Apply lockout/tagout per your facility's LOTO procedure.
3. Verify zero voltage at the terminal box using a calibrated tester.
4. Allow the motor to cool if it has been running (surface temperatures can exceed 80-90°C under load).
5. Do not open the terminal box or touch windings until isolation is confirmed.

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SECTION 3: NORMAL OPERATING INDICATORS

Parameter	Normal Range
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Bearing temperature	Up to ~70-80°C (alarm above ~90°C)
Winding temperature (Class F)	Up to ~155°C insulation limit
Vibration velocity (ISO 10816-3, small/medium machines)	Below 2.8 mm/s = good; 2.8-7.1 mm/s = acceptable; above 7.1 mm/s = unsatisfactory
Insulation resistance	Minimum 1 MΩ per kV of rated voltage, plus 1 MΩ (general industry rule of thumb), measured with a megohmmeter after isolation
Current draw	Should not exceed nameplate full-load current (FLA) under rated load

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SECTION 4: PREVENTIVE MAINTENANCE SCHEDULE

DAILY (visual, motor running):

- Listen for unusual noise (grinding, rattling, humming changes)
- Check for excessive vibration by feel/visual

- Check for overheating smell or discoloration
- Confirm no visible smoke, sparking, or fluid leaks at terminal box

WEEKLY:

- Check and clean cooling fan cover and vents (dust/debris blocks airflow)
- Check foundation bolts and coupling/belt alignment visually
- Record bearing temperature if a sensor/gun is available

MONTHLY:

- Measure vibration levels at DE and NDE bearing housings
- Check terminal box connections are tight (isolated motor only)
- Inspect drive belt tension/coupling condition (if applicable)

QUARTERLY:

- Measure insulation resistance (megger test), isolated motor only
- Re-grease bearings per lubrication schedule (Section 6)
- Check earthing/grounding continuity

ANNUALLY:

- Full insulation resistance and polarization index test
- Bearing replacement inspection (check for wear, noise, play)
- Full electrical connection torque check
- Complete vibration analysis and trend comparison

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SECTION 5: COMMON FAULT INDICATIONS (ERROR CODES)

These codes are used by this project's monitoring/error-code database.

Code	Description	Severity	Typical Cause(s)
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M-E01	Overcurrent / overload trip	High	Mechanical overload,
			bearing seizure, low
			voltage, phase imbalance
M-E02	High winding temperature	High	Blocked ventilation,
			overload, poor cooling,
			ambient temp too high
M-E03	High bearing temperature	Medium	Insufficient/excess
			lubrication, bearing wear,
			misalignment
M-E04	Excessive vibration	Medium	Misalignment, imbalance,
			loose foundation bolts,
			bearing wear
M-E05	Low insulation resistance	High	Moisture ingress,
			winding degradation,
			contamination
M-E06	Phase failure / single-phasing	High	Blown fuse, loose
			connection, supply fault
M-E07	Unusual noise (grinding/whining)	Medium	Bearing damage, rotor
			rub, foreign object
M-E08	Motor fails to start	High	Blown fuse, open winding,
			control circuit fault,
			mechanical jam

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SECTION 6: TROUBLESHOOTING PROCEDURES

M-E01 — Overcurrent / Overload Trip

1. Isolate and lock out the motor before physical inspection.
2. Check driven equipment for mechanical binding or jam.
3. Verify supply voltage is within $\pm 10\%$ of nameplate rating.
4. Check for phase imbalance (>2% imbalance can cause overheating).

5. If no mechanical/electrical fault found, escalate to a qualified electrician for insulation and load testing.

M-E02 — High Winding Temperature

1. Isolate and allow motor to cool before inspection.
2. Check cooling fan and vents are clear of dust/debris.
3. Confirm ambient temperature is within motor's rated range.
4. Check for sustained overload conditions on the driven equipment.
5. If temperature remains high after clearing airflow, escalate.

M-E03 — High Bearing Temperature

1. Isolate motor and allow to cool.
2. Check lubrication level and grease condition/color.
3. Check for shaft misalignment with driven equipment.
4. Listen for bearing noise after restart (grinding = likely wear).
5. If bearing noise persists, schedule bearing replacement.

M-E04 — Excessive Vibration

1. Isolate motor.
2. Check foundation bolts for looseness.
3. Check coupling/belt alignment.
4. Inspect for debris or imbalance on fan/coupling.
5. If vibration exceeds 7.1 mm/s (ISO 10816-3) after checks, escalate for professional vibration analysis.

M-E05 — Low Insulation Resistance

1. Isolate motor completely.
2. Inspect terminal box for moisture or contamination.
3. Clean and dry terminal box if moisture is found; re-test after drying.
4. If insulation resistance remains below the minimum threshold (Section 3), escalate — do not re-energize.

M-E06 — Phase Failure / Single-Phasing

1. Isolate motor.
2. Check fuses/breakers for all three phases.
3. Check terminal connections for looseness or corrosion.
4. If supply-side fault is suspected, escalate to facility electrician.

M-E07 — Unusual Noise

1. Isolate motor.
2. Rotate shaft by hand (if safe) to feel for roughness/binding.
3. Inspect for foreign objects near fan cover.
4. If grinding/whining persists, bearing replacement is likely required.

M-E08 — Motor Fails to Start

1. Isolate and check fuses/breaker/control circuit.
2. Check for blown fuse or tripped overload relay.
3. Attempt manual shaft rotation to rule out mechanical jam.
4. If electrical supply and mechanical rotation are both normal but motor still fails to start, escalate to qualified technician — possible winding fault.

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SECTION 7: LUBRICATION GUIDELINES

- Use grease type as specified on motor nameplate/lubrication tag.
- Do not mix incompatible grease types.
- Typical re-greasing interval: every 2,000-4,000 operating hours for standard industrial duty, or every 6-12 months, whichever comes first (actual interval depends on duty cycle, ambient temperature, and bearing type — always confirm with equipment-specific documentation).
- Over-greasing is a common cause of bearing overheating; follow specified grease quantity, do not "grease until it purges" excessively.

SECTION 8: SAFETY WARNING & ESCALATION

This document provides decision-support guidance only. If a fault cannot be resolved using the steps above, or if there is any doubt about electrical or mechanical safety, escalate immediately to a qualified maintenance technician or electrician. Do not attempt repairs beyond your training or facility authorization.

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